

IGARKA, Russian Federation

WMO: 232740

Lat: 67.4423N

Lon: 86.6053E

Elev: 92

StdP: 14.65

Time Zone: 7.00 (E07)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-52.1	-48.4	-56.4	0.2	-51.8	-53.0	0.2	-48.1	21.5	3.4	19.8	-0.4	1.4	330	0.791

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	79.6	64.0	75.5	62.6	71.3	60.6	66.5	75.6	64.2	72.0	61.9	69.7	6.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.1	86.8	70.6	60.8	79.8	69.1	58.4	73.2	65.9	31.3	75.8	29.5	72.0	27.8	69.9	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.3	17.0	15.4	-54.4	84.8	6.3	3.9	-59.0	87.7	-62.7	89.9	-66.2	92.1	-70.8	95.0		
			-54.5	69.7	6.0	2.7	-58.8	71.6	-62.3	73.1	-65.7	74.6	-70.1	76.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.4	-17.9	-14.3	1.6	15.5	30.8	53.6	60.2	53.1	42.0	22.7	-4.8	-12.0	(6)
	DBStd	30.64	18.53	16.01	16.49	13.92	10.20	9.69	7.47	6.79	7.71	11.94	18.22	18.42	(7)
	HDD50	11841	2104	1800	1502	1036	601	74	7	42	262	847	1644	1922	(8)
	HDD65	16705	2569	2220	1967	1486	1060	359	188	372	691	1312	2094	2387	(9)
	CDD50	671	0	0	0	0	6	181	323	140	21	0	0	0	(10)
	CDD65	58	0	0	0	0	0	15	39	4	0	0	0	0	(11)
	CDH74	520	0	0	0	0	3	143	350	24	0	0	0	0	(12)
	CDH80	96	0	0	0	0	0	20	75	1	0	0	0	0	(13)
	WSAvg	7.2	7.1	6.6	7.1	8.0	7.8	7.4	6.6	6.6	7.1	7.6	6.5	7.4	(14)
(15) Precipitation	PrecAvg	19.4	1.2	0.9	1.1	1.1	1.1	1.9	2.1	2.4	2.2	2.3	1.7	1.5	(15)
	PrecMax	25.2	2.5	1.9	1.9	3.0	2.2	4.3	5.6	4.6	5.2	3.9	2.9	2.5	(16)
	PrecMin	14.2	0.2	0.1	0.2	0.1	0.1	0.2	0.4	0.7	0.7	0.8	0.2	0.6	(17)
	PrecStd	3.1	0.6	0.4	0.4	0.6	0.5	0.9	1.3	1.0	0.9	0.7	0.6	0.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.1	24.4	35.2	48.6	67.6	82.8	85.9	77.4	68.4	49.7	33.9	26.5	(19)
		MCWB	20.3	23.8	31.6	39.6	55.1	62.7	68.6	65.8	58.4	47.4	33.0	25.5	(20)
	2%	DB	16.1	19.5	31.5	42.8	55.5	78.2	81.8	72.2	61.2	43.0	30.7	23.2	(21)
		MCWB	15.3	18.9	29.4	36.6	46.1	60.8	66.2	62.3	54.2	41.6	30.0	22.6	(22)
	5%	DB	10.8	14.6	28.4	38.5	49.2	73.6	77.8	68.0	56.9	39.2	26.5	18.4	(23)
		MCWB	10.2	13.8	26.7	34.4	41.9	60.0	64.2	60.1	52.6	37.8	25.8	17.6	(24)
	10%	DB	6.7	8.8	24.6	35.3	44.4	69.3	73.4	64.3	53.5	35.9	21.6	13.2	(25)
		MCWB	6.0	8.0	23.0	32.6	38.8	58.1	62.5	58.2	49.9	34.7	20.8	12.6	(26)
	0.4%	WB	20.3	23.8	32.2	40.7	54.7	66.3	70.6	68.4	59.3	47.4	33.0	25.7	(27)
		MCDB	21.0	24.3	34.1	46.8	65.8	77.3	82.7	74.4	66.0	49.9	33.3	26.4	(28)
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	15.2	18.7	29.5	37.3	46.6	63.4	67.9	63.8	55.9	41.8	29.8	22.4	(29)
		MCDB	16.0	19.4	31.3	41.8	53.8	73.7	77.9	69.3	60.1	43.0	30.6	23.0	(30)
	5%	WB	10.3	13.7	26.7	34.8	42.3	61.2	65.7	61.0	53.0	37.7	25.7	17.6	(31)
		MCDB	10.8	14.5	28.4	37.7	48.6	70.6	74.5	67.1	56.3	38.7	26.5	18.4	(32)
	10%	WB	5.9	8.0	23.0	32.5	39.1	59.1	63.7	58.7	50.0	34.8	20.8	12.7	(33)
		MCDB	6.5	8.7	24.5	35.2	44.3	68.0	71.4	63.8	52.9	35.7	21.7	13.3	(34)
	5% DB	MDBR	12.9	14.7	18.6	19.0	13.5	15.4	15.1	12.7	10.3	9.0	12.6	12.6	(35)
		MCDBR	14.5	17.1	14.3	14.7	17.2	21.9	22.0	18.0	16.3	8.7	11.1	13.8	(36)
		MCWBR	14.4	17.0	13.4	11.1	11.0	10.4	10.4	10.3	10.7	7.5	11.1	13.6	(37)
		MCWBR	14.5	17.5	14.4	13.6	16.5	19.3	18.9	15.4	13.8	8.0	11.1	13.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.220	0.263	0.311	0.333	0.359	0.377	0.410	0.396	0.335	0.268	0.203	N/A	(40)
		taud	1.857	1.979	2.001	2.050	2.129	2.378	2.282	2.285	2.406	2.137	1.886	N/A	(41)
	Ebn at Noon		30	164	222	255	262	262	246	235	225	165	31	0	(42)
		Edh at Noon	4	18	31	39	40	32	35	31	21	15	3	0	(43)
(44) All-Sky Solar Radiation	RadAvg		13	175	649	1301	1699	1719	1619	1004	540	193	36	0	(44)
	RadStd		2	19	72	125	153	146	201	117	63	34	6	0	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (3 neighbors)		+1.76	N/A	N/A	N/A	N/A	N/A	-670	-819	N/A	N/A	

Nomenclature: See separate page